A

## The selection problem defines biological and analytical axes

### INTRINSIC SUBSTRATE

Can this transcript form a suitable duplex?




sequence · predicted fold
genomic organization · burden

### CONTEXT / REALIZATION

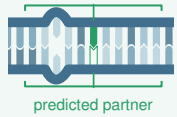
Does that substrate exist in the relevant state?




tissue variation · RBP / abundance
transcript / repeat · chromatin

### TARGET-LOCAL SELECTION

Which adenosine is selected within the duplex?




target geometry · focal state
aligned local neighborhood

**STATISTICAL DEPENDENCE** Sites within one duplex are dependent; label recovery and pair-first weighting were separated.

**AI / MODELING SEARCH** more supervision · cross-context sharing / multitask learning · richer representations · capacity / integration

B

## Controlled comparisons organize the retained evidence

| BIOLOGICAL SCALE | SEQUENCE–STRUCTURE REFERENCE | PAIR-DISJOINT DEVELOPMENT OOF | SAME-FOLD + MATCHED CONTROLS | EFFECT + UNCERTAINTY EVIDENCE STATUS |
|---|---|---|---|---|
| INTRINSIC SUBSTRATE | ● Sequence + predicted fold | ● Geometry → pair burden | × Broad repeat / transcript panel | **whole-duplex signal + between-substrate propensity** |
| CONTEXT / REALIZATION | ● Same-site tissue variation | × RBP × bulk abundance | ▲ Bulk H3K36me3 ○ Matched donor / cell in vivo state | **context dependence visible; responsible state unresolved** |
| TARGET-LOCAL SELECTION | ▲ Contact geometry + focal node | ● Aligned local representation | × Shifted matched-capacity control | **target-centered local signal recovered** |
| STATISTICAL DEPENDENCE | ● Additional valid high / low labels | ● Pair-first weighting | | **label recovery and weighting make separable gains** |
| AI / MODEL SUFFICIENCY | ▲ Graded objectives + shared tissues | × Foundation representations | × All-A + joint integration ● Fixed prediction integration | **targeted changes help; generic complexity yields little more** |

● supported / selected ▲ limited × unsupported formulation ○ matched data unavailable

C

## The completed search changes the next question

### RECOVERABLE

Whole-duplex sequence–structure
Pair geometry at substrate level
Aligned target-local information
Pair-aware statistical weighting

### PARTIALLY RESOLVED

Tissue-associated variation is visible
Bulk-context screens lack consistent gains
Tested model extensions yield limited returns

### MEASURE NEXT

Editing paired to donor and cell state
Transcript processing and regulator occupancy
ADAR state and in vivo RNA conformation

**FROZEN SYNTHESIS CHECK** The selected principles improved AUROC, AUPRC, F1, and Brier score in all five held-out endpoints.
unseen benchmark pairs · external-cohort transport remains untested